Observing the unobservable from animal movement data

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Introduction

Understanding animal movement and behaviour can aid spatial planning and inform effective conservation management. However, it is difficult to directly observe and record behaviours influencing animal movement in remote and hostile terrain such as the marine environment. As a result, the underlying behaviours often inferred from telemetry data using hidden Markov models (HMMs) are not typically validated. conservation actions drawn from wrongly inferred behaviours may be ineffective

Objectives

- Infer hidden behavioural states from movement data using HMMs.
- Validate inferred behavioural states using observed behavioural data.

Data

The Joint Nature Conservation Committee collected the data from 2009-2011 by visually tracking terns in some selected breeding colonies across the UK. The data consists of

- movement data of the boat used for the visual tracking
- behavioural data of terns recorded by observers on the boat

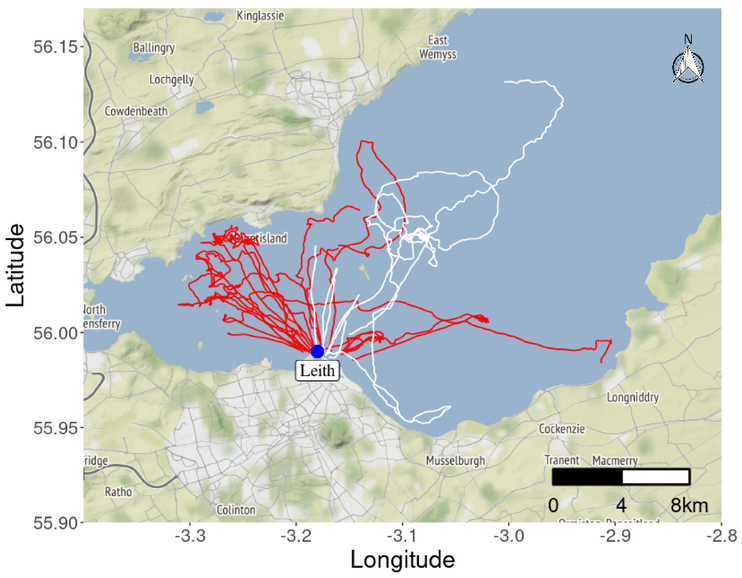

Figure 1. Visual tracks of common terns from Leith dock during incubation (white tracks) and chick-rearing (red tracks) seasons, 2010.

HMMs in movement ecology

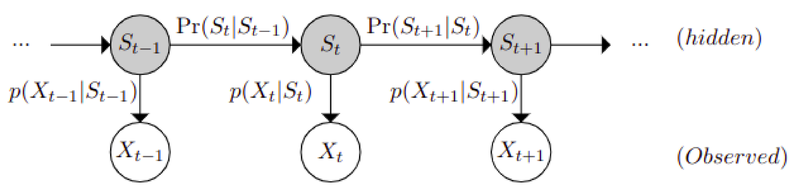

Figure 2. Graphical representation of a HMM where S_t denotes the hidden or state process and X_t denotes the observed or state-dependent process.

- S_t corresponds to the behavioural states of foraging ($S_t = 1$) and non-foraging ($S_t = 2$)
- X_t corresponds to the movement data converted into metrics of step length (km), l_t-gamma distribution, and turning angles (radians),ψ_t - von-Mises distribution
- $p_j(X_t \mid S_t) = p(l_t \mid s_t = j), p(\psi_t \mid s_t = j)$ for $j = 1, 2$

Result

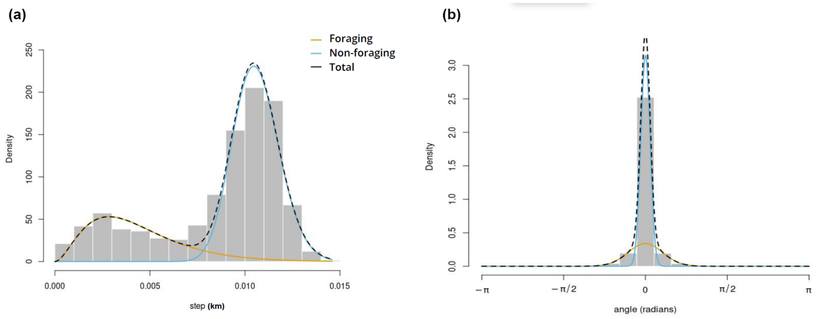

Figure 3. (a) Histograms of step length (km) and (b) turning angle of visually tracked common terns from South Shian colony during chick-rearing period, 2011. Lines represent HMM-fitted state-dependent distributions.

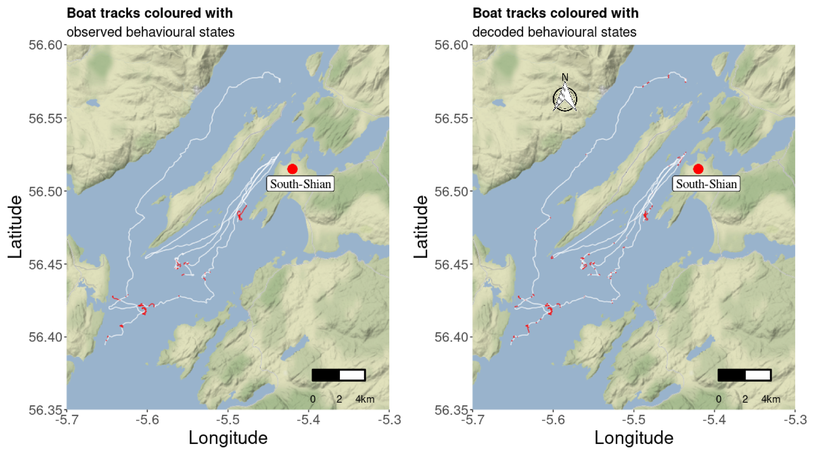

Figure 4. Visual tracks of common terns from South Shian colony coloured with observed (left) and inferred (right) behavioural states: foraging (red) and non-foraging (white).

Result

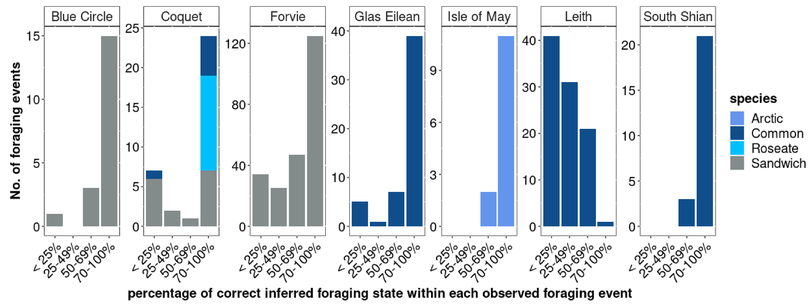

Figure 5. Proportion of correctly inferred foraging states within each observed foraging event of tern species across the study sites during chick-rearing. Inferred foraging states are from HMMs deemed optimal based on validation (i.e. HMMs with the least log-loss value).

Study Implication

- Using HMMs to infer important conservation-relevant behaviours from telemetry data is justified according to our validation results.
- Inferred behaviours from HMMs, particularly foraging behavioural activities, can inform the spatial planning and design of designated protected areas.

References

1. Langrock, R., King, R., Matthiopoulos, J., Thomas, L., Fortin, D. Morales, J. (2012),483 'Flexible and practical modeling of animal telemetry data: hidden Markov models and484 extensions.', Ecology 93, 2336–2342.

2. Zucchini, W., MacDonald, I. L. Langrock, R. (2016), Hidden Markov Models for Time547 Series: An Introduction Using R (2nd ed.)